

YOURSTORY



H2S



COCO CLI

HACKATHON

GCC Edition



Team Name : SnowCognition

Team Leader Name : Swati Gupta

Team Size : 3

Problem Statement : Patient and Member 360 & Clinical / Regulatory Document Copilot

Submission Guidelines:

(Things to add on the submission deck)

1. Problem Brief

- What real business problem does this solve?
- Who is the target user/persona?
- What is the current pain point and how does this improve it?
- Industry/domain context

2. Architecture Diagram

- System design showing data flow
- Which CoCo CLI skills are used and how they connect
- Data sources (structured/unstructured)
- How modular components plug together

3. Impact Statement

- Measurable outcomes (time saved, accuracy improvement, etc.)
- Scalability potential
- How this extends beyond the demo



Problem Brief



Real Business Problem

Care & life-sciences teams work across siloed EHR/claims data and dense unstructured documents (clinical notes, FDA guidelines). This leads to delayed decisions, missed safety signals, and regulatory non-compliance.



Target User / Persona

- Clinical Data Analysts reviewing patient risk portfolios
- Pharmacovigilance teams monitoring drug-safety guidelines
- Compliance officers cross-referencing regulatory documents



Current Pain Point → How SnowCognition Solves It

Pain: Copy-pasting between systems takes hours; insights are siloed and uncited.

Solution: One unified Streamlit dashboard on Snowflake — Patient 360 (structured) + AI Copilot that answers queries with CITED evidence from clinical notes & FDA guidelines.



Industry / Domain Context

Healthcare & Life Sciences | Regulatory Compliance | Clinical Decision Support
Built on 100% synthetic/de-identified data (HIPAA-safe demonstration).



Architecture Diagram

📁 Data Sources

- Synthetic Patients CSV
- Conditions / Medications
- Clinical Notes (TXT)
- FDA Guidelines (PDF)



Snowflake

- HEALTHCARE_COPILOT DB
- COPILOT_SCHEMA tables
- Internal Stage (docs)
- Cortex AI ready



Gemini 3.8 Flash

- Clinical Q&A
- Evidence retrieval
- Cited recommendations
- Regulatory lookup



Streamlit

- Patient 360 UI
- AI Copilot chat
- Risk colour codes



Structured Flow

CSV → generate_data.py → Snowflake Tables → SQL queries → Patient 360 Dashboard



Unstructured Flow

Docs → Snowflake Stage → Prompt context → Gemini AI → Cited Copilot answer



CoCo CLI Skills Used

- snowflake-connector-python — structured query skill
- google-generativeai — LLM inference skill
- streamlit — UI rendering skill
- pandas — data wrangling skill
- Streamlit Secrets — credential management



Modular Components: fetch_data() ↔ Snowflake | ask_gemini() ↔ Gemini API | Sidebar secrets ↔ Config | Fallback data ↔ Demo mode

 Measurable Outcomes

- Query time: 2–3 hrs → < 30 seconds per clinical question
- Evidence citation: 0% → 100% (every AI answer is sourced)
- Risk flagging: Manual → Automated colour-coded ( /  / )
- Regulatory lookup: Multi-day → Real-time in-context retrieval
- Data privacy: HIPAA-safe — 100% synthetic data used

 Scalability Potential

- Replace synthetic data with real EHR (Epic/Cerner) via Snowflake connectors
- Extend Gemini context with Snowflake Cortex Search over full document corpus
- Multi-tenant: each hospital/payer gets isolated Snowflake schema
- Add real-time streaming via Snowpipe for live patient monitoring

 Beyond the Demo — Real-World Extension

- Regulatory Compliance Automation: Auto-flag patients who violate FDA/guideline thresholds → reduce audit time by 60%
- Pharmacovigilance: Ingest MedWatch adverse event feeds into Snowflake Stage → Copilot monitors drug-safety signals continuously
- Clinical Trial Matching: Cross-reference Patient 360 profiles against trial eligibility criteria stored in Snowflake
- Multi-Modal: Extend Gemini to analyse medical imaging metadata alongside structured EHR data
- Enterprise: Deploy on Snowflake Native App framework for one-click installation by healthcare organisations



Live Demo & Repository



Live App (Streamlit Community Cloud):

<https://snowcognition-5pyisfhicsv9xeh4vymvrq.streamlit.app>



GitHub Repository (Public):

<https://github.com/swatiicfai/SnowCognition>



Team SnowCognition

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Challenge: Patient and Member 360 & Clinical / Regulatory Document Copilot



Stack: Snowflake · Streamlit · Gemini 3.8 Flash · Python · synthetic data

Thank You!

Team SnowCognition

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